

2. A truss is loaded as shown in Figure 2.

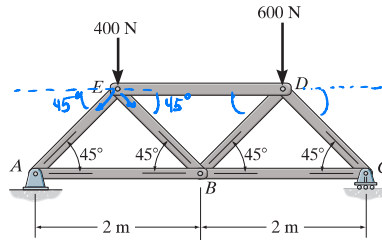


Figure 2: Truss loading

You are given the following sets of equations based on Method of Joints for joints A, B, and C.

$$A_x + AB + AE \cos(45) = 0$$

$$A_y + AE \sin(45) = 0$$

$$-AB + BC - BE \cos(45) + BD \cos(45) = 0$$

$$BE \sin(45) + BD \sin(45) = 0$$

$$-BC - CD \cos(45) = 0$$

$$C_y + CD \sin(45) = 0$$

$$\sum F_{x@E} = +ED - EA \cos 45 + EB \cos 45 = 0$$

$$\sum F_{y@E} = -400 - EA \sin 45 - EB \sin 45 = 0$$

D:

$$\sum F_x = -ED - DB \cos 45 + DC \cos 45 = 0$$

$$\sum F_y = -600 - DB \sin 45 - DC \sin 45$$

Assuming all members are in tension for sign consistency, write four additional equations using Method of Joints at joint D and E.

Solve for the 10 unknowns, AB , AE , BC , BD , BE , CD , DE , A_x , A_y , and C_y . For the sake of consistency (and my sanity for grading), let's all assume the same organization of our unknown vector:

3. A wall is constructed of concrete (grey) and insulation (blue) as shown in Figure 1. At equilibrium, the temperature profile (T_1 , T_2 , and T_3) in the wall is determined given T_0 and T_4 ambient air temperatures inside and outside the wall respectively. Heat transfer to the surfaces

eq 1:

$$\begin{bmatrix}
 AB & AE & BC & BD & BE & CD & DE & Ax & Ay & Ly \\
 1 & \cos 45 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
 0 & \sin 45 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
 -1 & 0 & 1 & \cos 45 & -\cos 45 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & \sin 45 & \sin 45 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & -1 & 0 & 0 & -\cos 45 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & \sin 45 & 0 & 0 & 0 & 1 \\
 0 & -\cos 45 & 0 & 0 & \cos 45 & 0 & 1 & 0 & 0 & 0 \\
 0 & -\sin 45 & 0 & 0 & -\sin 45 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & -\cos 45 & 0 & \cos 45 & -1 & 0 & 0 & 0 \\
 0 & 0 & 0 & -\sin 45 & 0 & -\sin 45 & 0 & 0 & 0 & 0
 \end{bmatrix}
 \begin{bmatrix}
 \bar{x} \\
 AB \\
 AE \\
 BC \\
 BD \\
 BE \\
 CD \\
 DE \\
 Ax \\
 Ay \\
 Ly
 \end{bmatrix}
 =
 \begin{bmatrix}
 0 \\
 0 \\
 0 \\
 0 \\
 0 \\
 0 \\
 0 \\
 0 \\
 400 \\
 0 \\
 600
 \end{bmatrix}$$

Verify solution:

@ D: $(-58.76) \quad (- \quad)$

$\Sigma F_y = -600 - BD \sin(45) - CD \sin(45)$

of the wall are governed by Newtons Law of Cooling: $q = h (T_f - T_i)$. Heat transfer through the wall materials are governed by Fouriers Law: $q = k/L (T_f - T_i)$.

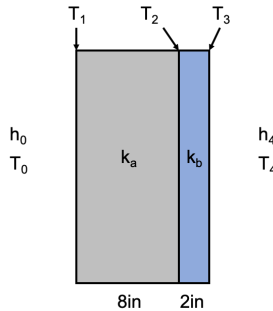


Figure 1: Composite slab subject to temperature boundary conditions
Write the 4 equations for heat transfer from the inside air into the concrete (q_1), through the concrete (q_2), through the insulation (q_3), and into the outside air (q_4):

$$q_1 = h_0 (T_0 - T_1) \quad q_4 = h_4 (T_3 - T_4)$$

$$q_2 = \frac{k_a}{8 \text{ in}} (T_1 - T_2)$$

$$q_3 = \frac{k_b}{2 \text{ in}} (T_2 - T_3)$$

To solve the system of equations, each heat transfer must equal according to conservation of energy $q_1 = q_2 = q_3 = q_4$, and $q_3 = q_4$. Express these 3 equations in matrix form:

$$q_1 = q_2 \quad q_2 = q_3 \quad q_3 = q_4$$

$$h_0 (T_0 - T_1) = \frac{k_a}{8 \text{ in}} (T_1 - T_2)$$

$$\frac{k_a}{8 \text{ in}} (T_1 - T_2) = \frac{k_b}{2 \text{ in}} (T_2 - T_3)$$

$$\frac{k_b}{2 \text{ in}} (T_2 - T_3) = h_4 (T_3 - T_4)$$

$$h_0 T_0 - h_0 T_1 = \frac{k_a T_1}{8 \text{ in}} - \frac{k_a T_2}{8 \text{ in}}$$

$$\frac{k_a T_1}{8 \text{ in}} - \frac{k_a T_2}{8 \text{ in}} = \frac{k_b T_2}{2 \text{ in}} - \frac{k_b T_3}{2 \text{ in}}$$

$$\frac{k_b T_2}{2 \text{ in}} - \frac{k_b T_3}{2 \text{ in}} = h_4 T_3 - h_4 T_4$$

$$h_0 T_0 - T_1 (h_0 + \frac{k_a}{8 \text{ in}}) + T_2 \frac{k_a}{8 \text{ in}} = 0$$

$$T_1 (\frac{k_a}{8 \text{ in}}) + T_2 (-\frac{k_a}{8 \text{ in}} - \frac{k_b}{2 \text{ in}}) + T_3 (\frac{k_b}{2 \text{ in}} + h_4) = 0$$

$$\begin{bmatrix} h_0 & -(h_0 + \frac{k_a}{8}) & \frac{k_a}{8} & 0 & 0 \\ 0 & \frac{k_a}{8} & -(\frac{k_a}{8} + \frac{k_b}{2}) & \frac{k_b}{2} & 0 \\ 0 & 0 & \frac{k_b}{2} & -(\frac{k_b}{2} + h_4) & h_4 \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix} = \begin{bmatrix} -h_0 T_0 \\ 0 \\ 0 \end{bmatrix}$$

Solve for unknowns T_1 , T_2 , and T_3 given $T_0 = 70^\circ \text{F}$, $T_4 = 10^\circ \text{F}$, $h_0 = 2 \text{ Btu}/(\text{hr} \cdot \text{ft}^2 \cdot ^\circ \text{F})$, $h_4 = 5 \text{ Btu}/(\text{hr} \cdot \text{ft}^2 \cdot ^\circ \text{F})$, $k_a = 0.3 \text{ Btu}/(\text{hr} \cdot \text{ft} \cdot ^\circ \text{F})$, $k_b = 0.02 \text{ Btu}/(\text{hr} \cdot \text{ft} \cdot ^\circ \text{F})$:

$$T_0 = 70^\circ \text{F} \quad h_0 = 2 \text{ Btu}/(\text{hr} \cdot \text{ft}^2 \cdot ^\circ \text{F})$$

$$T_4 = 10^\circ \text{F} \quad h_4 = 5 \text{ Btu}/(\text{hr} \cdot \text{ft}^2 \cdot ^\circ \text{F})$$

$$k_a = 0.3 \text{ Btu}/(\text{hr} \cdot \text{ft} \cdot ^\circ \text{F})$$

$$k_b = 0.02 \text{ Btu}/(\text{hr} \cdot \text{ft} \cdot ^\circ \text{F})$$

Find: T_1, T_2, T_3

Solution:

Rearrange matrix w/ givens

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$$T_0 = 70^\circ\text{F}$$

$$T_4 = 10^\circ\text{F}$$

$$h_0 = 2 \text{ Btu/hr-ft}^2\text{-}^\circ\text{F}$$

$$h_4 = 5 \text{ Btu/hr-ft}^2\text{-}^\circ\text{F}$$

$$K_A = 0.3 \text{ Btu/hr-ft}^2\text{-}^\circ\text{F}$$

$$K_B = 0.02$$

$$\begin{matrix} T_0 & T_1 & T_2 & T_3 & T_4 \\ \begin{bmatrix} h_0 & -(h_0 + \frac{K_A}{\delta}) & \frac{K_A}{\delta} & 0 & 0 \\ 0 & \frac{K_A}{\delta} & -(\frac{K_A}{\delta} + \frac{K_B}{t}) & \frac{K_B}{t} & 0 \\ 0 & 0 & \frac{K_B}{t} & -(\frac{K_B}{t} + h_4) & h_4 \end{bmatrix} \end{matrix}$$

$$140 \text{ Btu/hr-ft}^2 \begin{bmatrix} T_1 & T_2 & T_3 \\ -(\frac{h_0 + \frac{K_A}{\delta}}{\delta}) & \frac{K_A}{\delta} & 0 \\ \frac{K_A}{\delta} & -(\frac{K_A}{\delta} + \frac{K_B}{t}) & \frac{K_B}{t} \\ 0 & \frac{K_B}{t} & -(\frac{K_B}{t} + h_4) \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} -(\frac{h_0 + \frac{K_A}{\delta}}{\delta}) & \frac{K_A}{\delta} & 0 \\ \frac{K_A}{\delta} & -(\frac{K_A}{\delta} + \frac{K_B}{t}) & \frac{K_B}{t} \\ 0 & \frac{K_B}{t} & -(\frac{K_B}{t} + h_4) \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \end{bmatrix} = \begin{bmatrix} -140 \\ 0 \\ -50 \end{bmatrix} \text{ Btu/hr-ft}^2$$

Now into Python:

Results:

$$T_1 = 69.76^\circ\text{F}$$

$$T_2 = 57.10^\circ\text{F}$$

$$T_3 = 10.09^\circ\text{F}$$